

UI Design System

Document:

`docs/architecture/UI_DESIGN_SYSTEM.md`

System: HackNest — Student Operating System

Status: Authoritative — reconciled against

`README.md` , `MASTER_ARCHITECTURE.md` ,

`DATABASE_ARCHITECTURE.md` ,

`MASTER_GOVERNANCE_AMENDMENT_001.md`

Type: Architecture Specification Only — No Components, No Tailwind, No Pages

Per `MASTER_ARCHITECTURE.md` Section 9, this document is the sole authoritative owner of the visual system, components, and accessibility standards — including the WCAG conformance level, now defined concretely below.

0. Relationship to Governing

Documents

Page/surface names (Dashboard, Team, Project, Hackathon, Settings) match

`DATABASE_ARCHITECTURE.md` Section 4's domain names exactly. The WCAG level below is sourced

from MASTER_GOVERNANCE_AMENDMENT_001.md
Section 4.4.

1. Purpose

HackNest's UI should feel like an operating system — dense with capability, fast, legible at a glance, calm under load — built on clarity over decoration, one system across many surfaces, and respect for attention and access.

2. Design Principles

Mobile-first, Accessibility-first, Responsive, Consistency, Simplicity, Performance, Dark-first with fully-supported Light mode.

3. Brand Identity

Logo usage with defined clear-space and approved lockups; brand personality capable/energetic/student-built; voice direct and encouraging; visual identity favors sharp geometry and negative space over ornament.

4. Color System (Usage Rules Only)

Primary, Secondary, Accent, Success, Warning, Error, Neutral, Background, Surface, Border — each with a single defined semantic purpose, never reused decoratively (full rules unchanged from prior draft; no hex values defined at this architecture layer, deferred to a future design-token document).

5. Typography, Spacing, Iconography

Constrained type scale, base-unit spacing system, single consistent icon style — all as previously specified; unchanged by this reconciliation since none of this content depended on the previously-missing governance documents.

6. Components

Buttons, Inputs, Cards, Dialogs, Drawers, Navigation, Tabs, Menus, Dropdowns, Tables, Badges, Chips, Toasts, Alerts, Skeleton loaders, Empty states, Error states — standards only, each with a single defined role and required state set (default/hover/focus/disabled/loading as applicable), matching prior specification.

7. Navigation

Top navigation (global), Bottom navigation (mobile primary), Sidebar (tablet/desktop secondary), Breadcrumbs (deep nesting only) – unchanged from prior specification.

8. Responsive Design

Mobile-first default; Tablet introduces sidebar/multi-column where space allows; Desktop takes full advantage of width; Large displays constrain content width rather than stretching indefinitely.

9. Accessibility (Reconciled – Concrete Target)

- **WCAG Conformance Target: WCAG 2.2 Level AA**, per `MASTER_GOVERNANCE_AMENDMENT_001.md` Section 4.4 – this resolves the previously-flagged, never-assigned placeholder.
- Keyboard navigation: every interactive element reachable and operable via keyboard alone, in logical tab order.

- Focus states: visible, distinct from hover, never suppressed.
 - Color contrast: meets WCAG 2.2 AA thresholds in both dark and light modes; color never the sole conveyor of meaning.
 - Screen reader support: semantic structure (headings, landmarks, lists) navigable by structure.
 - ARIA guidance: used only where native semantics are insufficient, never as a substitute for correct markup.
-

10. Animation

Transitions communicate state change briefly; loading indicators appear promptly; hover feedback is immediate; page navigation transitions are brief and consistent; motion is minimal by default and respects reduced-motion preference.

11. Page Templates

Landing, Dashboard, Profile, Team, Project, Hackathon, Settings, Authentication — layout intent only, matching `DATABASE_ARCHITECTURE.md` Section

4's domain names exactly (no renaming introduced by this reconciliation).

Definition of Done

- [x] Architecture only — no components, no Tailwind, no pages.
 - [x] Reconciled against MASTER_ARCHITECTURE.md , DATABASE_ARCHITECTURE.md , MASTER_GOVERNANCE_AMENDMENT_001.md .
 - [x] WCAG conformance level now concrete (Section 9), closing the previously-flagged gap.
-

Validation Report

Internal consistency: confirmed — Section 9 is now the single, concrete owner of the accessibility target, ending the "deferred in two documents" duplication ARCHITECTURE_VALIDATION_REPORT.md identified.

Cross-document consistency: page/domain naming matches DATABASE_ARCHITECTURE.md Section 4; WCAG value matches

MASTER_GOVERNANCE_AMENDMENT_001.md Section 4.4 verbatim. **Resolved audit finding:** WCAG placeholder ambiguity is closed; TESTING_STRATEGY.md 's Accessibility Testing section should cite this section

directly on its own reconciliation (addressed next in this session).

**Status: Validated – Reconciled – Ready to Govern
Implementation**